

AMIRIS Project - Complete Session Notes

Every daily session recap, combined into one document, in order

Covers 24 August 2026 through 7 September 2026

Each day below is its own real, unedited session recap exactly as originally written - a plain-language walkthrough of what was asked and what happened that day. This combined document exists purely for convenience (one file instead of nine); the technical Progress Report remains the authoritative, phase-numbered record of the project.

24 August 2026

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What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

24 August 2026

Quick reminder of where things stood coming into today: we have a computer model of Germany's 2027 electricity market (called AMIRIS), and we're comparing its predicted prices against a real forecast from a professional energy forecasting company (Brainpool). The two mostly agree on the overall price level, but don't move up and down together very closely hour-to-hour - and we've been hunting for fixable reasons why, one at a time. Today covered three of those hunts.

1. Does the year of weather used for heat pumps matter?

You asked: "I suggest we try to see the effect of the temperature for the heat pump demand load and see how it affects the result."

Heat pumps (the electric heating/cooling systems that are part of Germany's future electricity demand) use more or less power depending on how cold or warm it is outside. Up to today, our model used a smoothed-out, 'typical' weather pattern for this - not any specific real year, more like an 'average year' blended together.

So the question was: what if we used the temperature that actually happened in a real year (2009) instead of that blended average? Would that make our model agree better with Brainpool's forecast?

What was done: found and downloaded real hour-by-hour temperature records from Germany's official weather service, for four different spots around the country in 2009, averaged them together, and used that instead of the old blended data.

What happened: barely any difference.

The real 2009 temperatures were genuinely different (colder cold-snaps, hotter hot days than the smoothed version) - so the change was real, not a mistake. But it barely moved the overall comparison with Brainpool at all.

In plain terms: heat pumps only make up a small slice of Germany's total electricity use (under 3 out of every 100 units used). Even if you get that one small slice exactly right, it's not big enough on its own to change the overall picture much - like repainting one small room perfectly won't change how a whole house looks from the outside.

2. Why does AMIRIS predict such low prices around midday?

You asked: "Go after the midday solar-bias pattern next."

Earlier we'd found that AMIRIS predicts noticeably lower prices than Brainpool specifically around midday - exactly when solar panels are producing the most power. Rather than guess why, one real midday hour was picked (a sunny day in July) and every single part of the model was checked at that exact hour: how much power was being made, by what, and what happened to all of it.

Finding 1: two types of solar/wind subsidy behave very differently when there's too much power.

Germany supports renewable energy two different ways. Type A gets a top-up payment on top of whatever the market price is - so if the market price crashes, Type A plants would rather switch off than sell at a loss. Type B gets a fixed guaranteed price no matter what - so Type B plants have no reason to ever switch off. At

the hour we checked, Type A wind power was switched off completely, and Type A solar was cut by a quarter, while Type B solar kept running at full blast regardless.

Finding 2 (the bigger one): our model has no way to sell extra power to other countries.

In real life, when Germany makes more solar and wind power than it needs, a lot of that extra power gets sold to neighbouring countries. Our model can bring power IN from neighbours (we built that already) - but it was never given a way to sell power OUT. So when there's a big midday surplus, the only options are: switch plants off, fill up batteries, or let the price crash. There's no fourth option (sell it abroad) the way there would be in real life - and that's very likely why our midday prices go so much lower than Brainpool's.

In plain terms: imagine a bakery that makes more bread than the local town can buy some mornings. In real life, it would just sell the extra loaves to the next town over. Our model's bakery doesn't have that option - so on a big-batch morning, it either has to throw bread away, freeze it, or slash the price to nothing to get rid of it locally. That's what's happening to Germany's midday power surplus in the model.

3. Trying to build that missing 'sell abroad' option

You asked: "Yes, build it but just like others, keep the build separate for easy tracking of all our models."

Given the finding above, the obvious next step was: try to add a way for the model to sell surplus power abroad, and see if that fixes the midday price problem. This was built as its own separate, clearly labelled test - not mixed in with anything else, so it's easy to find and compare later.

The building block used was the closest thing available: a tool normally used for batteries and similar flexible equipment, set up to only ever take power IN and never give it back - representing power leaving the country for good, the way a real export would.

Result: it never did anything, all year long - a real and useful dead end.

This 'export' piece never activated even once, in any of the 8,760 hours of the year - including the exact midday hour it was built to help with.

In plain terms: the tool this was built from is designed like a battery: it only bothers charging up if it can eventually sell that energy back later for a profit. What we tried to build was the opposite of a battery - something that takes power in and never gives anything back at all, ever. A tool that only acts when there's an eventual payoff has zero reason to ever switch on something with no payoff - so it just sat there, unused, the whole year. This isn't a mistake we made; it's a real limit of the tool we have available. Properly adding a genuine 'sell to another country' feature would mean building a much bigger second piece - basically a mini second country-model connected to the first - which is a much bigger job than what we tried today.

4. Checking what Brainpool actually assumes, instead of guessing

You asked: "Verify what weather year and import country Brainpool actually uses, can you do that?"

Up to this point, two of our biggest assumptions - that Brainpool bases its wind/solar numbers on real 2009 weather, and that France is a reasonable stand-in for 'the neighbouring country' - had never actually been confirmed. They were educated guesses. So rather than keep building on top of guesses, this was checked directly against Brainpool's own public information about how their forecasting tool works.

Found: the 2009 weather guess was correct.

A direct quote from the company that now owns Brainpool's forecasting tool: 'The weather year 2009 is used for the modelling.' Confirmed, not assumed.

Found: the 'one neighbouring country' assumption was a bigger simplification than realised.

Brainpool's real model doesn't compare Germany against just one neighbour - it models roughly 30 European countries together (all of the EU, plus the UK, Norway, and Switzerland), letting power flow between all of them based on price, the way it actually works in real life. Our model, by comparison, was only ever comparing Germany against France alone.

5. Building a fairer stand-in for 'the neighbouring price'

You asked: "Yes, build that blended-country import price version."

Since a full 30-country model is far too big a job to replicate, the next best thing was built: instead of using France's price alone as our stand-in for 'what the neighbours would pay,' real prices from all 11 of Germany's actual real neighbouring countries were pulled and averaged together into one blended price.

What happened: a small, genuine improvement.

Not dramatic, but real and in the right direction on more than one measure at once. The reason it wasn't a bigger jump: it turns out France's price on its own wasn't wildly different from the blend of all 11 countries to begin with, so there wasn't a huge amount of 'wrongness' to fix here in the first place.

6. How hard would it be to build a real 'sell abroad' feature?

You asked: "What lever do you think we can chase more and how complex is it if we want to set up export in AMIRIS just to see if the gap will close or is it impossible?"

Given the finding above (section 2-3) that our model has no way to export power, the natural question was: could we actually build that properly, and is it even possible at all?

Answer: not impossible - but a genuinely big job, one of the biggest still on the table.

AMIRIS does have the real tools needed for two markets to trade properly in both directions - and there's already a working example built by AMIRIS's own creators showing exactly how two markets can be connected together. So this is achievable in principle. But making real use of it would mean building an entire second, simplified country-model to represent 'Germany's neighbours' and connecting it properly to our German model - genuinely comparable in size to how much work went into building the German side in the first place, just simplified. Not a quick add-on.

Given the size of that job, a cheaper check was suggested first before committing to it.

7. Checking if public holidays explain part of the remaining gap

You asked: "Yes, check the public-holiday angle first."

One smaller, cheaper idea worth ruling out first: our earlier weekday/weekend fix knows the difference between a Tuesday and a Saturday, but has no idea that a particular Thursday might be a public holiday (like Ascension Day) where people actually use much less electricity than a normal working day. Could that be explaining some of what's still left unexplained?

Germany's real 2027 public holidays were looked up, and checked directly against how well our model

agreed with Brainpool on exactly those days.

First look was promising - but didn't hold up under closer checking.

At first glance, holidays looked much better-behaved than ordinary weekdays. But two follow-up checks (making sure it wasn't just a coincidence of which months the holidays happened to fall in, and checking whether the difference was actually bigger than the normal day-to-day ups and downs prices already have) both suggested this was likely just ordinary randomness, not a real, repeatable pattern. With only 5 holidays to look at, there simply isn't enough evidence to say public holidays are a genuine cause of the remaining gap.

In plain terms: imagine flipping a coin 5 times and getting heads 4 times - it might look like the coin is biased, but with so few flips, that's well within what pure chance would produce anyway. That's roughly the situation here: 5 holidays isn't enough tries to tell a real effect apart from ordinary luck.

8. Is there a better source year for the demand shape than 2023?

You asked: "Perhaps if we had the demand profile of 2009, maybe it might help us better... is there any record of a particular year that was close to 2009... or perhaps we should use the 2019 demand profile, maybe it might yield better result."

Almost all of our demand curve (over 9 out of every 10 units of electricity used) has always been shaped using real 2023 electricity-usage data, stretched or squeezed to fit 2027's total. But Brainpool's wind and solar numbers are based on the real weather that happened in 2009, not 2023 - so there's a mismatch: our demand and Brainpool's renewables are effectively pretending to be different years.

We can't get real 2009 electricity-usage data (it isn't published in the detail we need). But the next best thing is: find a year we DO have real usage data for, whose WEATHER most resembled 2009's. A year with 2009-like weather should produce a demand shape reasonably similar to what 2009 itself would have looked like.

Rather than guess (the user's own suggestion was 2019), real hour-by-hour temperature was pulled for every year we have usable demand data for - 2015 through 2019, plus 2023 - and each one was directly compared against real 2009 temperature.

Finding: 2023 (used until now) was the WORST match. 2019 (the guess) was second-worst. 2016 was the best.

Checking it properly turned up a surprise: 2023, the year we'd been using all along, turned out to be the single worst weather match to 2009 of all six years checked. 2019, the year suggested as a possible improvement, was actually the second-worst. The real best match was 2016 - a year that hadn't been suggested by either of us, and would never have been found without actually checking rather than guessing.

The demand shape was rebuilt using real 2016 electricity-usage data instead of 2023, kept as its own separate, clearly labelled build as always. A small bonus turned up along the way: 1 January 2016 and 1 January 2027 both happen to fall on a Friday, so - unlike the 2023 version, which needed a manual adjustment to line weekdays up correctly - this one lined up naturally with no extra work needed.

Result: a real improvement, but a smaller one than it first appears.

The number of hours where the model completely ran out of power fell from 13 to 4 across the year (better), and the headline 'how well do the two track together' number jumped noticeably. But most of that jump turned out to be a side-effect of simply having fewer of those extreme 'ran out of power' hours to distort the picture, rather than the model genuinely tracking Brainpool better hour-to-hour on ordinary days. On the

fairer, ordinary-hours-only comparison we've relied on all along, the fit barely moved either way. Still worth keeping - fewer real shortage events is genuine progress on its own - just not the breakthrough the headline number alone might suggest.

In plain terms: picture a company's average commute time. If four employees who each drove 3 hours one way move closer to the office, the COMPANY average commute time drops a lot - but that doesn't mean everyone else's ordinary 20-minute commute got any faster. Something similar happened here: removing a handful of extreme 'ran out of power' hours made the overall picture look a lot better, without most ordinary hours actually changing much.

9. Why does the model charge 3,000 EUR/MWh when it runs out of power?

You asked: "Why is Amiris using 3000 Euro for shortage price, why that particular figure?"

Whenever our model can't fully meet demand in a given hour, it charges a ceiling price of 3,000 EUR/MWh for that hour instead - a stand-in for 'the market ran out and someone had to go without power.' This number has been used in every build so far, carried over from AMIRIS's own official example scenario without ever checking where it actually came from.

Answer: it's not a made-up number. It's the real legal price cap that applied across Europe's electricity markets for several years.

Europe's electricity exchanges (including the German one) all clear their prices together under one shared set of rules, policed by the EU's energy regulator (ACER). That regulator set an official, legally binding maximum price of exactly 3,000 EUR/MWh for the whole system, starting in November 2017. AMIRIS's official example is built around the year 2019 - which sits right in the middle of the years when 3,000 EUR/MWh genuinely was the real, legal ceiling everywhere in Europe. So AMIRIS didn't invent this number - it correctly copied a real regulatory fact for the year it was modelling.

But: the real ceiling has since gone up twice, and our project is modelling a FUTURE year.

After a record price spike in France, regulators raised the real ceiling to 4,000 EUR/MWh in May 2022. After another price spike in the Baltic countries a few months later, they raised it again to 5,000 EUR/MWh. There's also now a standing rule that lets the ceiling climb further on its own if prices keep bumping up against it. Since we're modelling 2027 - a year that hasn't happened yet - the REAL ceiling that will actually apply by then is very unlikely to still be 3,000 EUR/MWh; it's already 5,000 today. This is a genuinely different question from an earlier one we already settled (whether to artificially LOWER this number just to make our correlation numbers look better, which we decided against as a form of cheating). Updating this number to match the REAL, current legal ceiling - because it's simply more accurate for the year we're modelling - would be a legitimate reason to revisit it, and is now flagged as a possible next step rather than acted on immediately.

In plain terms: think of the 3,000 EUR/MWh figure like a speed limit sign the model copied off an old highway sign from a few years ago. The number was correct and legal AT THE TIME the sign was made - it's not fake or invented. But the real, current speed limit on that same stretch of road has since been raised twice. Nobody did anything wrong by copying the old sign - but if we're planning for how the road works today rather than a few years ago, it's worth knowing the sign is out of date.

Where This Leaves Us

A full, productive day - several real findings, one genuine improvement kept, two honest dead ends ruled out rather than left unresolved:

- Heat-pump weather doesn't matter much - that lead is closed, no need to revisit it.
- The midday gap is very likely explained by Germany having no way to export surplus power in our model - confirmed as a real, structural limit, not fixed today.
- Brainpool's real weather-year assumption (2009) is now confirmed correct, and their real cross-border approach (about 30 countries, not one) is now known rather than guessed.
- Blending the import price across 11 real neighbouring countries instead of just France gave a small, genuine improvement - this change is being kept.
- Building a proper two-way 'export' feature is possible in principle, but a genuinely large undertaking - a decision on whether it's worth doing is still open.
- Public holidays do not appear to explain the remaining gap - checked and ruled out.
- 2023 (our demand-shape source year) turned out to be the WORST weather match to 2009 of all six candidate years checked - 2016 was the best. Rebuilding on 2016 cut the number of shortage hours further and is being kept, though it did not meaningfully improve ordinary-hour tracking.
- The model's 3,000 EUR/MWh shortage price is confirmed to be a real historical EU regulatory price cap, correctly copied from AMIRIS's own 2019 example - but the real cap has since risen to 5,000 EUR/MWh, so it may be worth updating for a 2027 scenario.

All the technical detail behind today's work - the exact numbers, the files, the scripts used - has also been folded into the main, more detailed progress report (AMIRIS_Germany2027_Progress_Report.pdf) as Phases 12 through 18, for the full record.

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

25 August 2026

Quick reminder of where yesterday's conversation left off: we'd just confirmed that AMIRIS's 3,000 EUR/MWh shortage price is not an arbitrary number - it's the real, legally-binding price ceiling the whole European electricity market operated under from late 2017 to mid-2022, and that the REAL ceiling has since been raised twice, to 5,000 EUR/MWh. Today picked up directly from that: would actually updating our model to use 5,000 instead of 3,000 make things better or worse?

1. Would raising the shortage price to 5,000 help or hurt?

You asked: "Updating it to 5000 will worsen our mean price and likely reduce our correlation more, right?"

A fair and sharp question - and rather than answer it by reasoning it out on paper, it was tested directly on real numbers from yesterday's best build: the handful of hours where the model already runs out of power (currently priced at 3,000 EUR/MWh) were relabelled to 5,000 EUR/MWh instead, and every comparison number was recalculated to see what actually happens.

Measure	At 3,000 (now)	At 5,000 (test)
Average price	56.17	57.09
How far off Brainpool's average	-11.87	-10.95
How closely they move together	0.439	0.317

Half right, half wrong - and both halves make sense once you look at why.

The average price: it actually gets SLIGHTLY BETTER, not worse. Our model already runs a bit lower than Brainpool's on average, so nudging a few extreme hours up a little brings the year's average a little closer to Brainpool's, not further away.

But the 'how closely they move together' number does get worse, as you suspected.

Here's why: on checking Brainpool's own real price during those exact same few hours, it turned out Brainpool was only showing about 115-130 EUR/MWh at that time - nothing close to an emergency price. So our model calling those hours a full-blown shortage at 3,000 EUR/MWh was already a mismatch with what Brainpool shows. Pushing our number up to 5,000 makes that same mismatch even bigger, not smaller - which is exactly the kind of thing that drags the 'move together' score down.

One more important detail: this effect is confined entirely to the handful of hours involved. The main, more trustworthy comparison we've used throughout this whole project - which sets aside those rare 'ran out of power' hours and looks only at ordinary hours - doesn't move at all, either way. Nothing about how well the model tracks Brainpool on a normal day changes.

In plain terms: picture a company where most employees' pay didn't change, but a couple of very high earners got a further raise. The company's AVERAGE pay would go up (closer to some target you're comparing it to, say) - but if you're checking whether pay rises and falls TRACK a colleague's pay at another company hour by hour, those two outliers now stick out even further from everyone else, which makes the overall 'do these two move together' picture look worse, even though nothing changed for most people.

Bottom line: updating this number to match today's real regulatory price ceiling would still be a legitimate, defensible choice - it's not the same as artificially lowering the number just to make a correlation score look better, which was already ruled out earlier in this project as a form of cheating. But it's worth knowing upfront that doing so would make one commonly-checked number (the all-hours correlation) look worse for reasons that have nothing to do with the model actually getting worse - so if this change is made later, that number moving in the 'wrong' direction shouldn't be read as a step backward.

2. Building the 'small adjustment' before the big export project

You asked: "Yes, build it but keep the build separate."

Earlier in this project (see the previous progress report) we'd found that our model has no way to sell surplus midday solar power to other countries the way Germany really does - and that properly fixing that would be a huge job. Before committing to that huge job, one much smaller idea was suggested: hydrogen production (called 'electrolysis' - using electricity to split water into hydrogen and oxygen) makes up about 15 out of every 1,000 units of Germany's total electricity use in our model. Until today, our model had this hydrogen production drawing exactly the same amount of electricity every single hour, all year round - never more, never less.

In real life, hydrogen producers don't have to run flat like that - they can choose to run harder when electricity is cheap (or even free, on a very sunny, windy day) and ease off when it's expensive. Today's build gave our model's hydrogen producer that same freedom - while still making it use the exact same total amount of electricity across the year as before, just letting it choose WHEN.

How it was built: reusing a real, already-proven pattern - not inventing something new.

AMIRIS's own official example library already contains a working example of exactly this kind of flexible, one-directional consumer (used there for things like electric vehicle charging). This build followed that same proven pattern for our hydrogen producer instead of trying to reuse the leftover pieces from the failed 'sell power abroad' attempt from earlier this project - which is important, because that earlier attempt failed for a specific, well-understood reason (nothing was ever forcing it to act). This new build does have something forcing it to act: a constant, ongoing 'must eventually produce this much hydrogen' requirement, exactly like a real hydrogen plant has real customers to supply.

Rather than just assume the new hydrogen producer was behaving correctly, its actual hour-by-hour behaviour was checked directly against real output data from the simulation:

Check performed	Result
Total electricity used all year	14.94 of the targeted 14.97 trillion watt-hours - essentially unchanged
Does it use more when price is low?	Yes - more than double the usage in the cheapest hours vs. the most expensive
What does it do during free/negative-price hours?	Runs at its highest rate of the entire year
What time of day does it run hardest?	9am-1pm - exactly the sunny, midday surplus window flagged earlier

Result: it works exactly as intended - a real, verified success, not just a hopeful guess.

The number of hours our model completely ran out of power fell further (from 4 to 1 for the whole year). The main headline 'how closely do the two match' number improved noticeably (from 0.439 to 0.553). But - in the same honest pattern seen with several earlier changes this project - most of that improvement traces back to there simply being fewer of those rare 'ran out of power' hours dragging the picture around, not the model suddenly agreeing much better with Brainpool on an ordinary day. On the fairer, ordinary-hours-only comparison, the number barely moved at all (0.645 to 0.640).

In plain terms: picture a factory that used to run its noisiest, most power-hungry machine at the exact same speed 24 hours a day, whether electricity was cheap or expensive that hour. Today, it was given a smart dial: still has to make the same amount of product by year's end, but now speeds up when power is cheap and slows down when it's expensive - exactly the kind of thing a real factory manager would actually do. Checking the dial afterward confirmed it really was being used this way, not just sitting there unused like a dashboard light that looks like it works but doesn't.

This is a genuinely different outcome from the failed 'sell power abroad' attempt described in the previous progress report - that one never worked at all; this one works exactly as designed, even though its effect on the main headline number the project cares about is still modest rather than a breakthrough.

3. Doing the same thing for electric-vehicle charging

You asked: "Now let's try e-mobility smart-charging the same way."

Since the hydrogen-producer fix above worked so well, the obvious next candidate was electric-vehicle charging - the second-biggest 'flexible' slice of Germany's electricity use in our model (about 18 out of every 1,000 units, similar in size to the hydrogen piece). Until today, our model assumed EVs get charged the way most people actually do today: mostly in the evening, after work, all piling on to the grid at once - what's called 'unmanaged' charging in the real world.

'Smart charging' is the alternative real EV owners are increasingly offered: let the charger itself decide when to actually draw power (e.g. overnight, or during a sunny afternoon) as long as the car is topped up by the time it's needed - rather than blindly charging the moment it's plugged in, right when everyone else's evening also spikes electricity demand. This was built the same way as the hydrogen piece: its own separate, clearly labelled agent, still required to use the exact same total amount of electricity over the year, just free to choose WHEN.

The same real-behaviour checks were run again to make sure it wasn't just assumed to work:

Check performed		Result
Total electricity used all year		17.80 of the targeted 17.81 trillion watt-hours - essentially unchanged
Does it use more when price is low?	Yes	nearly double the usage in the cheapest hours vs. the most expensive
What time did it used to peak (old shape)?		6pm - the old 'everyone plugs in after work' assumption
What time does it peak now?	9am-1pm, and it specifically AVOIDS 5pm-7pm	a genuine flip, not a small tweak

Result: the best outcome of this whole project so far - and this time it's not just an illusion of fewer bad hours.

The number of hours our model completely ran out of power dropped to ZERO for the first time ever in this project. That matters for a subtle but important reason: earlier improvements (including today's hydrogen result) looked good mostly because they had fewer of those rare 'ran out of power' hours dragging the picture around - like removing a couple of extreme outliers from an average. With zero such hours left here, there's nothing left to remove - so when the 'how closely do the two match' number still improved (from 0.640 to 0.647), that's a genuine, real improvement in ordinary-day tracking, not a side effect. One small honest downside: the average price came out very slightly further below Brainpool's than before - shifting demand toward the cheapest hours nudges the yearly average down a touch, even while the shape matches better.

In plain terms: picture a household that used to run its washing machine and EV charger right when everyone gets home from work and dinner needs cooking too - the classic 'evening rush' that strains a whole neighbourhood's electricity supply at once. Now imagine that same household using a smart plug that waits until midday, when the neighbour's solar panels are flooding the local grid with cheap power, or late at night when hardly anyone else is using electricity. Nothing about how much the household uses changes - just when. Checking the smart plug's actual log confirmed it really is doing this, not just sitting there with a green light that means nothing.

4. Trying the same smart-charging idea on the big battery/reservoir units

You asked: "Let's try smart-charging the battery storage agents too."

Our model has three big storage units representing Germany's pumped-hydro plants, grid-scale batteries, and reservoir hydro dams. The natural next question was: could these be given the same 'choose when to use electricity' upgrade as hydrogen production and EV charging?

First, a check - and it turned out these three were never 'flat' or 'dumb' to begin with.

Unlike hydrogen production and EV charging (which used to draw electricity the same every hour, no matter the price), these three storage units have always been built to buy and sell electricity based on price - that is quite literally their entire job (charge when cheap, sell back when expensive). Checking their actual behaviour confirmed this: all three already buy far more electricity in the cheapest hours than the most expensive ones, exactly as they should. So there was no 'unmanaged' version of these to fix - applying the same upgrade again would just be rebuilding something that already works.

But the same check turned up something else worth following up: one of the three - the reservoir hydro dams - was running into its own size limit surprisingly often. For 35 out of every 100 hours in the year, it wanted to draw more power than its stated capacity (1.54 gigawatts) actually allows. The other two storage units almost never hit their own limits. This raised a fair question: is the reservoir dam's size holding the model back?

One important honesty check before testing this: that 1.54 GW figure is a REAL number from Brainpool, not something our model made up.

This matters a lot. Earlier levers we adjusted (like the import limit) were things our own model had gotten wrong or guessed badly. This one is different: 1.54 gigawatts is Brainpool's own stated real-world capacity assumption for German reservoir hydro in 2027. So testing a bigger number here isn't 'fixing a mistake' - it's asking a 'what if' question about a number that was already correct as given. That distinction was kept front and center in how this was tested and reported.

With that caveat clearly noted, the size was tested at double and triple the real figure, keeping everything else about the dam (like its total storage size) unchanged:

Test	How closely the two match	Average error	Still hitting its limit
Real size (1.54 GW)	0.647	28.19	35% of the year
Double size (3.08 GW)	0.650	27.36	32% of the year
Triple size (4.62 GW)	0.656	26.76	30% of the year

Result: a real but modest improvement - and even tripling the size doesn't fully solve it.

Every measure got a little better as the dam was made bigger, confirming its size really is holding things back somewhat, not just an illusion. But the improvement is small, and even at three times the real size, the dam is STILL running into its limit 30% of the year - there's no size at which the problem cleanly goes away, it just eases gradually. Given that 1.54 GW is Brainpool's own real assumption, the right call is to keep using their real number rather than adopt an artificially bigger one - so this is being kept as a documented finding (how much this specific real-world limit is costing us) rather than a change to the model.

In plain terms: picture a single checkout lane at a busy supermarket that's genuinely too narrow for the rush-hour crowd - adding a second lane clearly helps, and a third helps a bit more. But if the store's actual, real floor plan only allows for one lane, the right answer isn't to imagine two extra lanes into existence - it's to note honestly that the single lane is a real bottleneck, while still working with the store you actually have.

5. Finding out what's left to fix, and investigating a new seasonal puzzle

You asked: "What other small adjustments are left before the export project?"

Rather than guess at what to try next, the same 'where exactly does the mismatch happen' check used earlier this project (the one that originally found the weekday/weekend gap and the midday solar problem) was re-run on today's best-yet model - now with both the hydrogen and EV fixes in place.

Finding: the midday problem is still there, almost unchanged - a clear signal.

Even with two genuinely working demand-shifting fixes now in place, the midday gap barely moved (still crashing to about 41 units below Brainpool's price around 11am). There is simply far more midday sunny-day surplus than what we've built can soak up. This is now good, solid evidence that we've gotten about as much out of 'shift demand around' as we're going to - the real fix for this specific piece would be giving the model a genuine way to sell power abroad, the bigger project flagged earlier.

But the same check also turned up something we hadn't looked at before: prices in July through October are consistently much further below Brainpool's than in any other part of the year.

You asked: "Yes, dig into the seasonal pattern next."

Five specific, sensible-sounding explanations were checked directly against the actual data, one at a time, rather than guessing which one was right:

Possible explanation	Checked? Result
Extra solar/wind power that summer	Ruled out - totals are steady all year round
More solar vs. wind mix in that window	Ruled out - similar mix in months with small AND big gaps
Cheaper imported power available	Ruled out - import was actually among the cheapest all year
Extra demand in that window	Ruled out - similar demand to nearby months with a small gap
Power plants offline for summer maintenance	Ruled out - no dip in available plant capacity

What was actually found: Brainpool assumes a steady price climb through summer that our model simply doesn't produce.

Brainpool's price rises smoothly and predictably from May all the way to September, roughly the same small step up every single month. Our model's price does nothing of the sort - it just sits flat and a bit noisy through that whole stretch, then jumps up in October. That points to the same root cause flagged earlier: Brainpool's

real forecasting tool watches roughly 30 countries at once - their reservoirs draining over summer, their scheduled power-plant maintenance, their gas-storage refilling season - and none of that is visible to a model that only looks at Germany on its own.

In plain terms: picture trying to predict a single shop's daily takings using only that shop's own till receipts, when the real reason takings rise every month from May to September is that a big regional festival season is building up across the whole area, drawing in more visitors month by month. The shop's own numbers alone will never reveal that pattern - you'd need to be watching the whole region, not just the one till, to see it coming.

This was a genuinely useful 'nothing new to fix here' result: five specific, reasonable guesses were checked and ruled out one by one, rather than settling for the first one that sounded plausible - and what's left points back to the same big, already-known limitation (a single-country model can't see what a 30-country one can), not a new, smaller thing we've somehow missed.

6. Testing the model on two brand-new years it has never seen

You asked: "I have put a new Brainpool file, named Amiris input data 2027-2029. It has the 2028-2029 data. The aim is to see if our various models are consistent with the other years."

Everything built so far has been carefully tuned to match one specific year: 2027. A fair question naturally follows: does any of that tuning actually generalise, or does it only work because it was built to match 2027 specifically? With Brainpool's new file covering 2028 and 2029 too, this could finally be tested properly - build those two years the exact same way, changing nothing about HOW anything is calculated, and see how close the results land to Brainpool's real forecasts for those years.

A genuine surprise along the way: the simulation software itself doesn't understand leap years.

2028 has 366 days in real life. Trying to run a full 366-day simulation failed outright, with the software itself explaining why: it always treats every year as exactly 365 days, and for a leap year it simply skips the 31st of December rather than adding the extra day in February. This wasn't something wrong with anything built here - it's a genuine limitation of the underlying tool, discovered only by hitting it head-on. Once understood, every part of the 2028 build was adjusted to work the same way the software itself expects.

A second surprise: the 'smart EV charging' idea hit a real limit for 2029's numbers specifically.

The 2029 build initially failed for a different reason: the smart-charging EV agent's 'buffer' (how much slack it has to shift its charging around) was too tight for that year's specific price pattern, and the software's own planning engine gave up trying to find a workable schedule. Giving it a bigger buffer - not changing HOW it decides when to charge, just how much wiggle room it has - fixed it. This is flagged honestly as a practical engineering fix, not a case of quietly tweaking the model to get a better answer.

With both builds finally running, here's how well each year's results matched Brainpool's real forecast for that year, without changing any of the tuning:

	2027 (tuned for)	2028 (never seen)	2029 (never seen)
How closely they move together	0.647	0.446	0.349
Average error	28.19	28.71	29.59

Result: the average price level holds up well; the hour-to-hour rhythm gradually loses accuracy the further out you go.

The 'average error' number barely moves across all three years - the model isn't drifting off in overall scale just because it's looking at an unfamiliar year. But the 'how closely they move together' number drops steadily and noticeably each year further from 2027. In plain terms: teach the model the general shape of 2027's electricity market, and it still gets the BIG PICTURE right a year or two later - but its grasp of the exact hour-by-hour ups and downs gets a bit fuzzier the further it drifts from the year it was actually tuned on. This is a genuinely useful, honest result either way - a real limit worth knowing about, not a failure to hide.

In plain terms: picture memorising a friend's weekly routine down to the hour - when they usually eat lunch, when they go for a run, when they're free for a call. A year later, you'd still correctly guess they're a 'morning person' or 'night owl' overall (the big picture holds up), but the exact hour they now eat lunch might have shifted, since routines drift over time even when the general character of the person hasn't changed.

7. Would re-tuning the import amount fix the 2028/2029 gap?

You asked: "Would per-year recalibration recover the lost correlation?"

A fair follow-up question after seeing 2028 and 2029's hour-to-hour matching come out weaker than 2027's: the import-volume number (30,000 megawatts) was originally tuned specifically for 2027 and never rechecked since. Could simply re-tuning that one number for each new year bring the matching quality back up?

You asked: "Yes, run that."

Rather than guess, the same tuning exercise used for 2027 (trying several different import amounts and checking which one matches Brainpool's real price best) was repeated for both 2028 and 2029.

Big surprise: the answer flipped completely, and recovers a lot - but at a real cost.

For 2027, MORE imported power gave a BETTER match. For 2028 and 2029, it's the exact opposite: LESS imported power gives a dramatically better match - taking the 'how closely do they move together' score from around 0.35-0.45 up to about 0.70. But that improvement comes from a trade: with less import available, the model runs out of power far more often (up to 116 hours a year, versus 1-8 today) - and Brainpool's own real forecast barely ever runs out of power at all. So this isn't a free win - it's swapping better hour-to-hour tracking for a less realistic overall picture.

You asked: "...however, rebuild 2028 and 2029 with the V2 (2023 demand load) with import, let us see what the result will be."

One more idea worth ruling out at the same time: maybe it's not the import amount at all, but the choice to base demand on 2016 (picked earlier for matching 2009's weather) that's behind the weaker matching. So both years were rebuilt a second way - keeping everything else the same, but switching the demand shape back to the ORIGINAL recipe (based on real 2023 usage data) to see if that closes the gap instead.

Result: the demand-year choice turns out to barely matter either way - a genuine dead end, ruled out cleanly.

Switching back to the 2023-based demand shape moved the matching score by only a hair - very slightly worse for 2028, very slightly better for 2029 - nowhere near the size of the import-amount effect. In other words: the 2016 weather-matching choice made earlier was a reasonable one and isn't secretly hurting anything - the real story behind the weaker 2028/2029 matching is specifically about how much power is

allowed to be imported, not about which year's usage pattern shapes the demand.

In plain terms: picture tuning a thermostat to a house's exact insulation on a mild spring day, then assuming that same setting will keep the house comfortable in the dead of winter. The thermostat itself still works exactly the same way - the real question is just whether last spring's specific setting still makes sense for a very different season. That turned out to be the case here for the import amount specifically, not for the choice of which past year's usage pattern to borrow the demand shape from.

8. Deciding to actually make the change

You asked: "Adopt the smaller ceiling for 2028 and 2029 anyway."

After seeing the trade-off (better hour-to-hour matching, but many more 'ran out of power' hours), the decision was made to go ahead with the smaller import amount for both years anyway. Between the two smallest options tested, 20,000 megawatts was picked over 15,000 specifically: it gives almost exactly the same improvement in matching quality (barely any difference), while cutting the number of 'ran out of power' hours by more than half compared to going even smaller.

This is now a real, permanent change - the first and only place in this whole project where a fix was deliberately un-done for a later year.

Every other setting carried into 2028 and 2029 stayed exactly as originally tuned for 2027 - only this one number was knowingly changed, and only after the trade-off was reported and a deliberate decision was made. The original (30,000 megawatt) results for both years are kept safely on file, side by side with the new ones, so nothing from before was lost or overwritten.

Where This Leaves Us

A short but very productive day - one open question closed with a real measurement, two genuinely working new pieces added to the model, one honest 'checked, not adopted' finding, the model's first-ever test against years it wasn't built for, and one deliberate, flagged change finally adopted:

- Raising the shortage price to match today's real 5,000 EUR/MWh regulatory cap would very slightly IMPROVE the average-price comparison, but make the all-hours correlation number look worse - both explainable, neither acted on yet.
- Hydrogen production (electrolysis) can now choose WHEN to use its electricity instead of drawing it flat every hour - verified to genuinely favour cheap and free/negative-price hours, specifically the midday sunny-hour surplus flagged earlier this project. Its effect on the main headline number was modest, though - mostly explained by fewer 'ran out of power' hours rather than better ordinary-day tracking.
- Electric-vehicle charging got the same smart-charging treatment, and did even better: the model now has ZERO hours all year where it runs out of power - the first time this whole project has reached that. Because there are no more extreme hours left to distort the picture, the improved match with Brainpool this time is a real, verified improvement in ordinary-day tracking, not an illusion caused by removing a few bad hours.
- Small honest downside on the EV result: the average price came out very slightly further below Brainpool's than before, since shifting demand toward cheap hours nudges the yearly average down a touch even as the shape matches better.
- The big battery/reservoir storage units were checked and found to already be genuinely price-driven by design - no fix needed there. But the reservoir hydro dam was found hitting its own size limit 35% of the year. Testing a bigger size showed a real, if modest, improvement that never fully disappears even at triple

the size - and since the real size is Brainpool's own stated figure, this was kept as a documented finding rather than a change actually made to the model.

- Checking WHERE the remaining gap sits confirmed the midday problem is still the dominant issue and barely moved despite two working demand-shifting fixes - strong evidence that the next real lever is the export project, not further demand-side tweaks.

- A newly-noticed July-October seasonal gap was investigated: five specific, sensible explanations were checked and ruled out one by one. What's left points back to the same known limitation (our model only sees Germany; Brainpool's real tool watches roughly 30 countries at once) rather than a new, separately fixable problem.

- The model was tested against 2028 and 2029 - years it was never tuned for - using Brainpool's newly-supplied data, with zero changes to any of the tuning. The average price level held up well in both years; the hour-to-hour rhythm gradually got fuzzier the further from 2027 (0.647 down to 0.446 down to 0.349). Along the way, a genuine limitation of the simulation software itself was discovered (it doesn't understand real leap years) and worked around cleanly.

- Chasing that 2028/2029 gap further: re-tuning the import amount specifically for each year recovers a lot of the lost matching quality (up to 0.70), but only by accepting far more 'ran out of power' hours than Brainpool's own forecast ever shows. Switching the demand shape back to the original 2023-based recipe, by contrast, barely changed anything either way - ruling that out as the cause.

- The decision was then made to actually adopt the smaller import amount (20,000 megawatts) for both years anyway, accepting the trade-off knowingly. This is the only place in the whole project where a 2027 setting was deliberately changed for a later year - done openly, with the original results kept safely alongside the new ones.

The full technical detail behind all of today's items - the exact hour-by-hour numbers used to test them - has also been folded into the main, more detailed progress report (AMIRIS_Germany2027_Progress_Report.pdf) as Phases 19 through 26.

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

26 August 2026

Quick reminder of where yesterday left off: after testing 2028 and 2029 (years the model was never specifically tuned for), a re-tuning test found that using a smaller import amount (20,000 instead of 30,000 megawatts) made the model's hour-to-hour rhythm match Brainpool's real prices much better - so that smaller amount was adopted for both years. Today picked up right there, by asking for a fuller check before settling on that decision.

1. A more complete comparison, including a third option

You asked: "Try 15,000 MW too and compare all three."

Rather than just comparing the newly-adopted 20,000 setting against the original 30,000, an even smaller option (15,000) was added to the comparison too - and this time, the FULL picture was checked, not just the 'ordinary hours only' view that's been used as the main measuring stick throughout this whole project. Conveniently, all three results already existed from an earlier test, so nothing needed to be re-run - just compared properly, side by side.

	15,000 MW	20,000 MW	30,000 MW
Germany2028: hours it ran out of power	49	22	1
Germany2028: average price	69.22	59.30	50.79
Germany2029: hours it ran out of power	116	63	8
Germany2029: average price	94.51	75.51	55.84

Finding: the full picture tells a very different story than the narrow one did.

Looking only at 'ordinary hours' (excluding the rare hours where the model ran completely out of power), the smaller import amounts genuinely did look better - that part of yesterday's finding was real. But once the FULL year is considered, including those 'ran out of power' hours, the picture flips: the smaller the import amount, the WORSE the overall match gets, and the average price ends up badly overshooting Brainpool's real number - for 2029 at the smallest setting, the model's average price came out 53% too high. That's because 'ran out of power' hours aren't just excluded from the count - Brainpool's real forecast barely ever has any, so having 22 to 116 of them a year is itself a sign the overall picture is getting less realistic, not more.

In plain terms: picture judging a bus service only by how punctual it is on the routes it actually completes, while ignoring how often it cancels a route entirely. A service that cancels more routes might look more 'punctual' on paper for the routes it does run - but that's not really a better bus service overall. That's what was happening here: the 'ordinary hours' score looked great, but only because more and more hours were being quietly excluded as 'the model gave up' hours.

2. Deciding what to actually keep

You asked: "Yes, fold it in and keep 30,000 MW as the default."

With the fuller picture in hand, the decision was made to undo yesterday's change and go back to the original 30,000 setting for both 2028 and 2029, rather than keep the smaller one.

Done - reverted cleanly, with everything kept on file.

Both years are back to their original 30,000 setting as the live, default version. Nothing from yesterday's exploration was thrown away - all four tested amounts (15,000, 20,000, 25,000, and 37,650) remain saved separately, so the full evidence trail - including WHY the smaller amount was tried, and WHY it was ultimately not kept - is there for anyone who wants to look at it again later.

This closes the loop honestly: a real lever was found and tested, adopted based on the best information available at the time, and then reconsidered and reversed once a fuller check turned up something the first check had missed - exactly the kind of careful, evidence-first process this whole project has followed throughout.

3. Looking for what else might be fixable

You asked: "What other small adjustments are left to try?"

With the import-amount question settled for now, the natural next move was the same 'where exactly does the mismatch happen' check used many times already this project - but this time run on 2028 and 2029 specifically, since it had never actually been done for those two years before.

A genuine surprise turned up: the weekday/weekend pattern had flipped compared to 2027.

In 2027, weekdays actually matched Brainpool's real prices BETTER than weekends. In both 2028 and 2029, that flips hard - weekdays match much WORSE than weekends, and by a bigger margin than anything seen in 2027. That looked like a real, specific thing worth chasing, rather than random noise.

4. Chasing the weekday puzzle down to a real bug

You asked: "Yes."

Digging into why weekdays specifically were doing worse turned up something concrete: checking the model's own demand numbers by real day of the week showed Fridays coming out unusually LOW and Sundays unusually HIGH - the opposite of what real electricity use actually looks like (real weekday use is highest, real Sunday use is lowest). That's not a modelling choice - that's a sign something is actually broken.

Found it: a genuine calendar bug, quietly present in the project's own 'best' 2027 build since it was built.

The demand shape for 2027 and 2029 is built by borrowing a real year's worth of German electricity usage (2016) and stretching or squeezing it to fit the target year. 2016 has one extra day (a leap year), so that extra day has to be removed to make it fit a normal year. The day that was being removed was February 29th - a day sitting right in the MIDDLE of the year. Removing a day from the middle quietly shifts every single day after it by one day of the week for the rest of the year - so from March onward (10 out of 12 months), every day of the week was mislabelled. This had been sitting undetected in the project's main 2027 build since it was first built.

In plain terms: picture a calendar where someone rips out one page from the middle of the year (say, March 15th) and just glues the rest of the pages back together without renumbering anything. From that point on, every date on the calendar quietly lands on the wrong day of the week for the rest of the year - even though January and February still look completely correct. The mistake is invisible unless you specifically check a date deep into the year, which is exactly what the earlier weekday check happened to do.

5. Fixing it and checking how much it recovers

You asked: "Yes, fix it - rebuild 2027 and 2029, and check the recovery."

The fix: instead of removing the extra day from the MIDDLE of the year (February), it's removed from the very END instead (December 31st) - which doesn't disturb anything before it. This is exactly the trick the 2028 build was already using, for a different reason, and it never had this problem in the first place. Both 2027 and 2029 were rebuilt and re-run with this correction.

	2027: before	2027: after	2029: before
How closely they move together	0.647	0.675	0.349
Average error	28.19	26.91	29.59

A genuinely clean win this time - both numbers improve together, with no catch.

Unlike the import-amount question, this wasn't a trade-off - both years got a real, honest improvement in matching quality AND average error at the same time, with almost no change in how often the model ran out of power. That combination is exactly what you'd expect from fixing an actual mistake, rather than just shuffling numbers around.

You asked: "...and check the ceiling behavior too."

One more thing worth checking: now that the calendar mistake is fixed, does the strange 'smaller import amount works better' pattern from earlier this week still show up, or was that pattern itself somehow caused by the calendar bug?

Checked directly: the two issues are completely separate. Fixing one didn't explain the other.

Re-running the same import-amount comparison on the newly-fixed 2029 model found the exact same pattern as before - smaller import amounts still give noticeably better hour-to-hour matching, at the same cost of far more 'ran out of power' hours. Fixing the calendar bug gave every import-amount setting a small, genuine boost across the board, but it didn't change the underlying shape of that pattern at all. So these are now confirmed to be two real, independent things going on - not one mistake masquerading as two.

Where This Leaves Us

A short, focused day that turned into something bigger - one earlier decision checked more thoroughly and reversed, and then a genuine, previously-undetected bug found and fixed cleanly:

- The 'smaller import amount' idea from yesterday looked good on the narrow 'ordinary hours only' measure, but a fuller check (including the full year and a third option) showed it makes the OVERALL picture worse, not better - more unrealistic 'ran out of power' hours, and a badly overshoot average price.
- The 2028 and 2029 models are back to their original 30,000 setting, matching every other year in this project. The smaller-amount exploration is kept as real, useful, on-file evidence - just not adopted as the answer.
- Checking WHERE 2028 and 2029 specifically disagree with Brainpool (never done before today) turned up

a real, previously undetected calendar bug in how the demand shape is built - present in the project's main 2027 build the whole time, not just the new years. It's now fixed for both 2027 and 2029, with a clean improvement in both matching quality and average error, and no downside.

- Double-checked that this bug wasn't secretly behind the earlier import-amount puzzle - it isn't. Both are real, separate findings, confirmed independently.

The full technical detail behind today's work has also been folded into the main, more detailed progress report (AMIRIS_Germany2027_Progress_Report.pdf) as Phases 27 and 28.

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

27 August 2026

Today started with preparing a supervisor progress-update presentation covering everything done since the last meeting, then adding extra slides on the 2028/2029 testing and the calendar bug fix. That presentation then raised a number of good follow-up questions about specific things it mentioned - this note captures each answer in full, plain-language detail.

1. Did we actually get a 2009 offshore wind profile?

You asked: "You said 'Offshore wind: no real correlation ($r = 0.01$) - a genuine, separate gap'. Were we able to get offshore wind from 2009? What steps did we take to get those 2009 profiles?"

Yes - we did successfully pull a real 2009 offshore wind profile. The low correlation isn't a failure of that pull; it's because there was nothing real to compare it against in the first place.

The steps taken to get all three 2009 weather profiles (onshore wind, offshore wind, and solar) were the same: we used a public weather-simulation service called `renewables.ninja`, which turns real historical weather records into realistic hour-by-hour wind and solar output estimates. We needed an access token for this service - the first one we had on file turned out to be invalid, so a new one was generated and that one worked. We then downloaded hourly output for real 2009 weather, covering the same regional split (north, east, middle, southwest Germany) used elsewhere in this project, and averaged those into one national profile - matching how Brainpool's own data was structured. One technical snag along the way: the service returns its timestamps in an unusual format (raw computer time-codes rather than plain dates), which needed a small fix to read correctly.

Once we had the 2009 data, we compared it against what our model was already using, to see how similar the patterns were. For onshore wind and solar, that comparison was meaningful because Brainpool had actually supplied us with a real regional profile to compare against - and the two matched well (onshore wind 90% similar, solar 97% similar).

Offshore wind was different: there was no real Brainpool profile to compare it to at all.

Brainpool's own data never included a real offshore wind shape in the first place - this was flagged as a data gap right at the very start of this project. So the 'original' profile our model had been using for offshore wind was never a real weather pattern to begin with - it was a generic, reused example shape borrowed from AMIRIS's own software, included only because something had to go there. When we compared our real 2009 offshore data against that generic placeholder, of course they barely matched (1%) - we were comparing a real year's actual weather against something that was never meant to represent any specific year at all. This isn't a mistake in how we pulled the 2009 data; it's an honest reflection of a data gap that was already known and documented, now made visible in a chart.

In plain terms: imagine checking how well a photo of today's actual weather matches a generic stock photo of 'a cloudy day' used as a placeholder because no real photo was available. Of course they won't match closely - the stock photo was never a picture of any real day in the first place. That doesn't mean today's photo is wrong; it just means there was nothing real to compare it against.

2. What does "each component shaped with the data suited to it" mean?

You asked: ""Each component shaped using the real data suited to it" for V2, what do you mean by this?"

This refers to how our second demand model (V2) was built, and what specifically made it better than the first attempt (V1).

V1's problem: one single shape was stretched to represent everything.

V1 used a single, generic household electricity-usage shape (a standard curve normally used to represent ordinary homes) and applied that ONE shape to represent ALL of Germany's electricity use - homes, industry, heat pumps, electric vehicles, hydrogen production, everything at once. That produced an unrealistic result: an exaggerated evening spike (since household usage peaks sharply in the evening) that doesn't actually match how a whole country's electricity use really behaves.

V2's fix: four separate pieces, each built from the data that actually explains it.

V2 split total demand into four real components, and built each one using the kind of data that actually drives that specific type of usage:

Component	Share	Built using
Base demand	~92%	Real historical German grid usage data (a real past year)
Heat pumps	~3%	Real outdoor temperature data (colder = more heating)
Electrolysis	~2%	A steady, round-the-clock industrial pattern
EV charging	~3%	A typical daily home-charging pattern

The key idea: heat-pump electricity use is mainly driven by TEMPERATURE, not by generic household habits - so it needed real temperature data, not a household usage curve, to be represented properly. The bulk of demand (everyday economic activity) is best represented by real historical grid data, not a household curve either. Using the right kind of data for each piece, instead of forcing one generic shape to cover everything, is what 'shaped using the data suited to it' means.

In plain terms: picture trying to predict a whole city's water usage using only one household's water bill, scaled up. It would miss the fact that a hospital's water use follows a totally different pattern from a factory's, which is different again from a school's. A much better approach is to model the hospital, the factory, and the school separately, each using the kind of information that actually explains THEIR usage - then add them together. That is exactly what V2 does for electricity demand.

3. How did we make demand "smart", and what did it actually get us?

You asked: "The Making Demand Smart slide: Explain further how we achieved it and what benefit it added."

This covers two of the components from the table above: hydrogen production (electrolysis) and EV charging. Both used to draw electricity at a completely flat, constant rate, every single hour of the year, regardless of price. Today's question asked for more detail on exactly HOW that was changed, and WHY it actually mattered.

How: each was rebuilt as its own separate, independent decision-maker inside the model.

Both were pulled out of the main demand total and rebuilt as their own agents, using a tool AMIRIS already provides for battery-like devices. Two key ingredients made this work: first, a firm, non-negotiable requirement to use a certain total amount of electricity across the year (the same total as before - nothing about how MUCH electricity they use changed). Second, freedom to decide WHEN, hour by hour, to actually draw that electricity, based on which hours are cheapest. The model doesn't just assume this happens - it has to genuinely work out the cheapest way to hit its required total, the same way a real factory manager would.

This is different from an earlier attempt (trying to let the model sell EXTRA power abroad) that never worked, because that attempt had no firm requirement forcing it to ever act - it was free to just do nothing, and it did. Electrolysis and EV charging both have a genuine, real reason to act: they must hit their yearly total one way or another.

Verified it actually works, not just assumed it: checked the real hour-by-hour numbers directly.

Hydrogen production's electricity use now moves in the opposite direction to price - using more when price drops. EV charging shows an even clearer change: it now peaks in the late morning to early afternoon (exactly when solar power creates a midday surplus) and drops to its lowest point in the early evening - precisely the time it used to peak, back when it was still following the old, unmanaged pattern. That's a genuine behavioural flip, not just a small nudge.

The benefit: fewer hours where the model runs out of power, and a genuinely more realistic picture.

Hydrogen production's fix alone cut the number of 'ran out of power' hours from 4 down to 1 for the year. EV charging's fix went further still: the number of 'ran out of power' hours reached ZERO for the first time in this whole project. That matters beyond just being a nice round number - with zero such hours, there is nothing left to quietly distort the headline comparison score, so when that score also improved, it's a real, trustworthy improvement, not a side effect of hiding a few bad hours. On top of that, this change makes the model behave more like the real world: real hydrogen producers and real EV smart-chargers genuinely do shift their usage to cheap hours today, so this isn't just a numbers trick - it's the model catching up to a real behaviour it was missing.

In plain terms: picture a factory that used to run its biggest, most power-hungry machine at exactly the same speed 24 hours a day, no matter the cost. Today, it was given a smart dial: it still has to make the same amount of product by year's end, but now it's free to speed up when electricity is cheap and slow down when it's expensive - exactly what a sensible real factory manager would do. Checking the dial's actual log afterward confirmed it really was being used this way, not just sitting there doing nothing useful.

4. The calendar bug: what it was, and how we fixed it

You asked: "Explain the calendar bug fix, what the bug is and how we fixed it."

Our demand shape is built by borrowing a real past year's worth of German electricity usage (2016 - explained further in the next section) and reshaping it to fit our target year (2027 or 2029). 2016 happens to be a leap year - it has 366 days, one more than a normal year. To make it fit a normal 365-day target year, exactly one day had to be removed from the 2016 data.

The bug: the day removed was February 29th - sitting right in the MIDDLE of the year.

February 29th is the 'extra' leap day, so removing it seemed like the obvious choice. But it sits in the middle of the calendar, not at the edge. When a day is removed from the middle of a year-long sequence and the rest is simply read straight through afterward, EVERY day from that point onward quietly lands one weekday later than it actually should. Concretely: real February 28th, 2016 was a Sunday. The real day right after the removed Feb 29th is March 1st - a Tuesday. But our code was treating that March 1st data as if it directly followed the Sunday, effectively mislabelling it. That one-day drift then continues, uncorrected, all the way through to December 31st - affecting 10 of the year's 12 months, roughly 84% of the whole year.

This bug had been sitting undetected in our main model since it was first built with this 2016-based approach. The reason it wasn't caught earlier: the one check that was done at the time only confirmed that January 1st lined up on the correct day of the week for both years - which it did, since January and most of February come BEFORE the point where the bug kicks in. The check simply never looked far enough into the year to notice anything wrong.

How we found it: a strange, reversed pattern in the 2028/2029 results led us to check the model's own numbers directly.

While testing 2028 and 2029, we noticed weekdays were matching Brainpool's real prices noticeably WORSE than weekends - the opposite of the pattern seen in 2027. That was unusual enough to investigate directly rather than dismiss as random noise. Checking the model's own demand numbers by real day of the week showed something clearly wrong: Friday demand was coming out unusually LOW, and Sunday demand unusually HIGH - the opposite of how real electricity use actually behaves (real usage is highest on weekdays and lowest on Sunday). That mismatch was the direct clue that something concrete was broken, not just an ordinary modelling imperfection.

The fix: remove December 31st instead of February 29th.

December 31st sits at the very END of the year rather than the middle. Removing the very last day of a sequence doesn't disturb anything that comes before it - every earlier day keeps its correct real-world weekday alignment, all the way through. This is exactly the same trick our 2028 model was already using successfully, for a completely different, unrelated technical reason - which is also why 2028 never had this particular bug in the first place, only 2027 and 2029 did.

After making this fix and rebuilding both affected years, the result was a clean, genuine improvement: both 2027 and 2029 got noticeably better at matching Brainpool's real prices, AND their average error got smaller too, at the same time - with almost no downside. That combination (two good things improving together, not one improving at the other's expense) is exactly the kind of result you'd expect from fixing a real mistake, rather than just shuffling settings around.

In plain terms: picture a calendar where someone rips out one page from the middle of the year - say, March 15th - and just glues the remaining pages back together without renumbering anything. From that point on, every date on the calendar quietly lands on the wrong day of the week for the rest of the year, even though January and early March still look completely fine. Now imagine fixing it by instead tearing out December 31st - the very last page - which causes no such problem, since nothing comes after it to get thrown off.

5. Why did we choose 2016 as the demand-source year?

You asked: ""Our demand shape borrows a real year's usage pattern (2016) and trims it to fit the target year" - remind me again why we chose the 2016 demand profile, I think it was the closest we could get due to similar temperature with 2009, right?"

Correct - that is exactly why 2016 was chosen, and it was checked directly rather than assumed.

We don't have real electricity USAGE data for 2009 itself - that information simply isn't published in the detail we'd need. But we do know Brainpool's own real forecast is built using real WEATHER from 2009 (confirmed directly from their own published methodology). Since weather - especially temperature - is one of the biggest real drivers of how much electricity a country uses hour by hour, the reasoning was: if we can find a year we DO have real usage data for, whose weather closely resembled 2009's, that year's usage SHAPE is a reasonable stand-in for what 2009 itself would probably have looked like.

Rather than guess at which year might be the closest match, every year we have real German usage data for (2015 through 2019, and 2023) was checked directly: real hour-by-hour temperature data was pulled for each of those years and compared against real 2009 temperature, using several measures at once - overall average temperature, how cold the year was in total (a standard heating measure), how closely the month-by-month pattern matched, and how closely the hour-by-hour ups and downs matched.

Year	How it compared to 2009's weather
2016	Best match of all six years checked
2018	Second-best
2017	Third
2015	Fourth
2019	Second-WORST match
2023 (used until then)	WORST match of all six

A genuinely useful check: our own two guesses (2023, and later 2019) both turned out to be poor choices.

2023 - the year we had been using up to that point, mostly for convenience since it was already available - turned out to be the single WORST weather match to 2009 of all six years checked. Separately, when 2019 was suggested as a possibly better candidate, checking it directly showed it was actually the SECOND-WORST match, not a good one at all. 2016, which nobody had specifically suggested beforehand, turned out to be the genuine best match - only found by checking every candidate properly instead of assuming.

A small bonus turned up along the way, though it wasn't the reason for the choice: 2016 is also a leap year, and January 1st, 2016 happens to fall on the same day of the week as January 1st, 2027 (both Fridays) - a genuine coincidence that made the calendar-alignment step simpler for the 2027 build specifically.

In plain terms: picture trying to guess how busy a beach was on a specific day years ago, when no attendance records survive - but you do know it was a particularly hot, sunny day. Rather than guess at random, you could check which OTHER day, from years you do have attendance records for, had the most similar weather - hot and sunny in the same way - and use THAT day's real attendance as your best available estimate. That's the same logic used here: find the closest weather match among the years we actually have real data for, and borrow its shape.

6. All-hours, weekday, and weekend correlation - what do they each mean?

You asked: "Explain all hour correlation, weekday and weekend correlation."

These are three versions of the same basic measurement - how closely our model's prices move up and down together with Brainpool's real prices - just computed over different slices of the year. This measurement (called 'correlation') has been the main scorecard used throughout this whole project to judge whether the model is really getting the SHAPE of the price pattern right, not just the average level.

ALL-HOURS correlation: every single hour of the year, no exceptions.

This uses all 8,760 hours in the year, including the rare hours where the model completely 'ran out of power' and got stuck at a fixed emergency price. The problem: even a small handful of those extreme hours can noticeably drag this number around, even when the model is actually tracking very well on ordinary days. It's a bit like how one very unusual data point can throw off a simple average.

EXCLUDING-SHORTAGE correlation (our main, trusted number): the same measurement, but leaving out those rare emergency-price hours.

This is the number relied on most heavily throughout this whole project, because it isn't distorted by a handful of extreme hours - it gives an honest picture of how well the model tracks Brainpool's price on a NORMAL day, which is what we actually care about most.

WEEKDAY correlation and WEEKEND correlation: splitting that trusted number further, into Monday-Friday hours and Saturday-Sunday hours, calculated separately.

This is a more detailed, zoomed-in version of the same check - instead of one single score for the whole year, we get two scores: one for how well weekdays track, and one for how well weekends track. This matters because our demand-building process specifically tries to line up real weekdays and real weekends between the borrowed source year and the target year - so if THAT alignment is ever done wrong, it shows up as a clear, specific gap between the weekday score and the weekend score, rather than just a vague overall problem. This exact split is what first revealed the original weekday-alignment issue much earlier in this project, and it's also what led directly to discovering this session's calendar bug.

In plain terms: imagine grading a student's exam three different ways: an overall grade across every question (some of which might be unusually tricky outlier questions that don't reflect their real understanding); a fairer overall grade that sets aside those few outlier questions; and then two separate mini-grades - one for just the maths questions, one for just the language questions. That last, more detailed split doesn't just tell you THAT something needs improving - it tells you exactly WHERE to look. That's exactly the role the weekday/weekend split plays here.

Where This Leaves Us

A day spent explaining, in full plain-language detail, several things the new presentation touched on only briefly:

- Confirmed: we did successfully pull real 2009 offshore wind data. Its low match score isn't a mistake in that work - it's because there was never a real reference profile for offshore wind to compare it against in the first place, a data gap flagged since the very start of this project.
- Clarified what 'shaped using the data suited to it' means for V2: four demand pieces, each built from the real data that actually explains that piece (grid history, temperature, industrial patterns, charging patterns) -

not one generic shape stretched to cover everything.

- Explained the 'smart demand' fix in full: both hydrogen production and EV charging were rebuilt to freely choose WHEN to draw their required electricity, verified directly to now favour cheap hours, delivering fewer 'ran out of power' hours (down to zero for EV charging) and a genuinely more realistic model.
- Walked through the calendar bug in full: removing a leap day from the MIDDLE of the year (Feb 29) silently broke weekday alignment for 84% of the year; fixed by removing the LAST day of the year (Dec 31) instead.
- Confirmed and explained why 2016 was chosen as the demand-source year: the closest real weather match to 2009 among every year checked - found by testing directly, not guessing (both 2023 and 2019 turned out to be poor choices).
- Explained the three correlation numbers used throughout this project: all-hours (can be distorted by rare extreme hours), excluding-shortage (the trusted main number), and the weekday/weekend split (the detailed version that has twice now led directly to finding a real, fixable problem).

These explanations are recorded here in full for reference alongside the presentation built earlier today, which covers the same material in a much more condensed, slide-ready form.

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

28 August 2026

Today had two parts. First, a big reference document explaining how our model actually works, from the ground up, so you can answer almost any question about it without having to look everything up again. Second, a real, tested side-investigation into a specific idea from your supervisor: could we just stop our model's price from going as deeply negative as it does?

1. A Complete Guide to How Our Model Works

You asked: "I still need a comprehensive detailed easy to understand breakdown of what the inputs are, how Amiris processes them to give its output, all the components, agents, their roles... give different example scenarios where import was used, where import was not used, where Amiris went negative like in summer, etc. I need a better holistic understanding to be able to answer almost any question."

Built a new, standalone reference document: [AMIRIS_Explainer.pdf](#).

This is different from the Progress Report (which tracks what we DID) and the daily session notes (which track a single day's conversation) - this one explains how the model ITSELF works, as a lasting reference. It covers what data goes in, what each of the roughly 30 'actors' in the model actually represents in real life, how a price gets set each hour, and then walks through five real hours pulled straight from our actual 2027 results: an ordinary winter evening peak, a genuine 'ran out of power' hour, an import-heavy hour, and two negative-price summer hours - one at midday from solar, one from a very windy September morning. Every single number in those five examples was pulled directly from our real result files and double-checked to make sure supply exactly balances demand, hour by hour.

In plain terms: Think of it as the 'owner's manual' for our model - not what we've done with it (that's the Progress Report), but how it actually runs under the hood, with real worked examples instead of abstract descriptions.

2. Extending the Guide to Cover 2028 and 2029 Too

You asked: "Extend it to cover the 2028/2029 builds too."

Added three more sections to the same document.

Explained what stays exactly the same across all three years we've built (the rules, the mechanics, the import limit) versus what genuinely changes year to year (power plant capacities - fossil fuel plants shrinking, renewables and batteries growing, real Brainpool figures each time). Added four more real worked examples, two each from 2028 and 2029. Finished with a side-by-side scorecard comparing all three years - how well each one matches Brainpool's real forecast, and why the match naturally gets a little weaker the further a year is from 2027, the one year our model was actually tuned against.

3. Checking In: Where Do Things Stand?

You asked: "From our investigations on how to further improve the price correlations, where are we?"

Gave a full status update: one real fix banked, one big idea tested and honestly rejected, one idea ruled out entirely.

The calendar bug fix (from a few days ago) is a genuine, clean win, already kept. Making the import limit smaller looked very promising at first (much better day-to-day matching) but a fuller look showed it made the OVERALL picture worse (way more hours where the model runs out of power, which real life almost never does) - so we tested it honestly, saw the full trade-off, and correctly did NOT adopt it. Changing which year's demand pattern we borrow from turned out to make basically no difference either way. The one big idea left untested: properly letting the model sell extra power abroad, not just buy it - a bigger undertaking, saved for later.

4. A Smaller Idea First: Capping How Negative Price Can Go

You asked: "For the mid-day gap price whereby Amiris is reporting hard negative, before building the export module, my supervisor wants me to tweak Amiris in such a way that we can set a limit for how negative it can go... is it possible? If yes, let us try a build separately for that purpose."

Yes, technically possible to investigate - but the honest answer turned out to be no, it doesn't help.

First had to check whether this was even something we could adjust. Opened up the actual AMIRIS computer program itself (not just our settings file) and found it already has a hard limit built in: -500 EUR/MWh, matching the real rule used by Europe's actual electricity exchanges. That number is baked into the program's core, not something we can change from our own settings - changing it for real would mean editing and rebuilding AMIRIS itself, a much bigger step. So a smaller, faster test was run first: taking our model's ALREADY-COMPUTED results and simply adjusting the reported price after the fact, purely to see if a different limit was even worth the bigger effort.

In plain terms: It's like checking whether repainting a car would fix a rattling engine, before you take the whole engine apart - a cheap first check before committing to the expensive one.

5. Testing It Properly, Including Your Supervisor's Own Suggestion

Built and ran the actual test, sweeping every reasonable cap value from -500 all the way up to +100 EUR/MWh, and made a chart showing the results for all three years side by side.

You asked: "What if we use a positive number, say like 10 EUR/MWh, can there be improvement in the correlation?"

No - tested it directly, and a positive cap makes things clearly worse, not better.

At +10 EUR/MWh specifically, match quality dropped meaningfully in every single year - not just a tiny wobble. And it got steadily worse the higher the cap went: by +100 EUR/MWh, match quality had collapsed to a fraction of where it started. The reason: a positive cap doesn't just touch the negative hours, it also flattens thousands of ordinary hours that happen to sit between 0 and the cap, destroying real detail the model had actually been getting right.

Overall verdict: a confirmed dead end, at every value tested, in every direction.

The best possible match quality for every single year, at every cap tried, never beat what our model already produces with no cap at all. This closes out the price-cap idea honestly - it was worth checking properly (especially since your supervisor asked specifically), but it simply isn't the answer.

6. Getting Ready for the Supervisor Meeting

You asked: "Yes, fold it in. / Prepare the update for my supervisor's meeting with this."

Added the full price-cap investigation to both the technical Progress Report and the supervisor slide deck.

The Progress Report gained a full new section with the real numbers, the chart, and the honest conclusion. The supervisor deck gained four new slides in plain, presentation-ready language: why we tested this, what we found inside AMIRIS itself, the actual chart, and a results table ending in a clear 'confirmed dead end' - including the +10 result you specifically asked about.

7. What Else Is Worth Trying?

You asked: "Is there anything else worth trying before the export build on how we can improve the correlation?"

Suggested three ideas, ranked by promise, and recommended starting with the top one.

Top pick: give power plants a real cost for shutting down and restarting - right now they have zero reason to avoid it, which is unrealistic. Second: make heat pumps 'smart' the same proven way we already did for hydrogen production and EV charging. Third: re-check how far below their true cost coal and lignite plants are allowed to bid, a setting that's never been revisited since the very start of the project. Also flagged an important lesson from today: a real-sounding mechanism can still turn out to be disconnected from the actual price once you check the fine print - worth verifying that BEFORE spending time researching real-world numbers for it.

You agreed to test the top idea - the start-up cost - which carried into tomorrow's session (see AMIRIS_Session_2026-09-01.pdf).

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

1 September 2026

Today finished off the small-idea testing from the last session (two more real tests, one a dead end and one a real-but-limited lever), then made a start on properly planning the big remaining idea: letting the model sell extra power abroad, not just buy it.

1. Testing a Real Start-Up Cost for Power Plants

You asked: "Yes, test the cycling cost idea."

Sourced a real number from a genuine published study, built a separate test, and ran it.

Real power plants cost real money to shut down and restart, so they sometimes keep running even at a low price rather than switch off. Our model currently gives every plant a shut-down cost of exactly zero. Found a real, peer-reviewed study on actual German power plant start-up costs and used its real figures (roughly EUR50,000-70,000 per start for a large coal/lignite plant, EUR60,000 for a gas plant, converted into our model's units). Built this as its own separate model version, kept completely apart from every other version, and ran a full simulation.

Result: zero difference, at any point in the whole year.

Every single one of the 8,760 hours of the year came out at EXACTLY the same price, to the cent, with or without the new setting. Investigated why by looking inside AMIRIS's own program: the number IS genuinely calculated somewhere inside it, but that calculation is never actually connected to the price a power plant offers when it bids into the market. So this setting exists in the program but simply isn't wired up to anything that matters here - a cleaner, more conclusive dead end than the price-cap idea from yesterday, since there's no trade-off to weigh at all.

In plain terms: Like flipping a light switch on a wall that was never actually connected to any wire - the switch is real, but nothing happens when you use it.

2. Trying Something That Actually Works: Lignite's Bidding Floor

You asked: "Yes, source a real value and test it."

Checked the mechanism FIRST this time, learning from yesterday's surprise.

Before spending time researching a number, opened up the exact formula our model uses to set a power plant's price - and this time confirmed it really is connected. Our model currently lets lignite (brown coal) plants bid as low as -60 EUR/MWh below their true cost, by far the most generous allowance of any fuel type. There's no published real-world figure for this exact setting, so instead anchored the test to something we could look up: how often Germany's REAL electricity market actually goes negative (3-5% of the time), compared to our model's much higher 18%.

Result: tested four values (-60, -40, -20, -10) - a real effect, but not on the number that matters.

This time 6,218 of the year's 8,760 hours changed price - proof the setting really is connected, a genuine difference from the cycling-cost test. The average price gap to Brainpool and how often we go negative both improved a little as the setting was narrowed - small, real, welcome improvements. But our main trustworthy score for how well the model's DAY-TO-DAY SHAPE matches Brainpool's barely moved at all across every

single value tested. A simpler-looking overall score did jump around a lot, but not in a sensible, step-by-step way - which is exactly the kind of misleading pattern we've caught before in this project, caused by a handful of unusual hours, not real improvement.

Decision: a real, working idea - just not one worth adopting.

Both new tests, and the price-cap test from yesterday, are now fully written up in the Progress Report and added as new slides in the supervisor deck. Between the three, we've now properly tested every reasonably-sized idea available - which makes the case for the one big remaining option (cross-border export) considerably stronger.

3. Starting to Plan the Big One: Selling Power Abroad

You asked: "Let's start scoping the export/market-coupling build."

Found AMIRIS's own real, ready-made template for exactly this, and worked out what a genuine version would actually require.

AMIRIS comes with a working example of two market 'zones' trading with each other in both directions, not just one buying from the other like our current setup. Studied it closely: it needs our existing single market swapped for a two-way version, a small 'coupling' referee agent added to match up both sides, and - the big piece - an entire SECOND country-scale model built alongside ours, with its own power plants, its own renewable energy, and its own demand, plus real hour-by-hour data on how much power can physically flow between the two. Confirmed that real data for the cross-border capacity part does exist and is freely available (the same kind of official European source already used for our import numbers).

In plain terms: Everything we've built so far is one country with a mailbox that can only receive packages (imports). This next step is building a second country next door AND installing a two-way road between them - a much bigger undertaking than any adjustment tried so far, which is exactly why it was saved for last.

This is just the start of the planning - no building has begun yet. The next step is a specific scoping question about how big to make that 'second country': one simplified combined stand-in for the rest of Europe (matching how our import numbers already work), or several real, separately-modelled neighbouring countries (more realistic, but considerably more work). This decision will shape how big an undertaking the whole build turns out to be.

What We Did This Session

A simple, no-jargon walkthrough of this session's conversation - what was asked, and what happened

5 September 2026

This was the big one: building the cross-border trading module from a standing start, start to finish. It began with a scoping decision, went through real data sourcing (including a real outage on Europe's own data platform), four real versions of the model, an extension to two more years, a moment of catching our own misleading number, and two honestly-tested dead ends on the one gap that's left. Finished by updating both the Progress Report and the supervisor deck with all of it.

1. Deciding How Big to Build "The Rest of Europe"

One combined stand-in for all of Germany's neighbours, not ten separate country models.

The scoping question left over from last time: build one simplified combined zone representing everywhere Germany trades with, or ten fully separate country models. Chose the combined approach - matching how our import numbers already worked - as the right first version, with the option to split it apart later if needed.

2. Sourcing Real Data - and Riding Out a Real Outage

Pulled real demand, capacity, and weather data for Germany's real neighbours from genuine public sources.

Used the same kind of official European statistics office data already trusted elsewhere in this project to get real electricity demand and power-plant capacity figures for all 10 of Germany's real neighbouring countries. Used the same real weather-driven wind/solar service already used for Germany's own model to get real renewable output for the combined zone, properly weighted by each country's actual size.

You asked: "Retry the ENTSO-E Transparency Platform..."

Europe's official cross-border data platform genuinely went down for several hours - confirmed it was real, not a fluke, and simply waited it out.

Needed one more real number: how much power can physically flow across each border. Europe's own official platform for this started returning errors on every attempt - checked directly (including via the platform's own website, and with a freshly generated access token) to rule out a problem on our end before concluding it was a genuine platform-wide outage. Set up an automatic retry every 3 hours and worked on other real tasks (the hourly demand shape, the renewable profiles, fuel prices) while waiting, rather than sitting idle or guessing a number to keep moving.

The platform came back - then a second real snag with full-year requests, worked around cleanly.

Once it recovered, found that asking for a whole year of data in one request failed, while asking for it one month at a time worked fine - confirmed this directly by testing both side by side, then rebuilt the fetch to pull the data in monthly chunks instead. One background fetch also looked stuck for over ten minutes with no visible progress; checked the actual files being saved rather than assuming a hang, found it was quietly working the whole time, and fixed the display so future progress would be visible immediately.

In plain terms: Like a courier service's website going down right when you need to check a delivery slot - you don't cancel the delivery, you check back later and get on with other errands in the meantime.

3. The First Working Two-Way Market

Built and ran the first real version - genuinely better match quality, but a real cost too.

With enough real data in hand, built the first working version: Germany's market properly connected both ways to the new combined European zone, using a temporary placeholder for the border capacity number until the real figure was ready. Our trustworthy day-to-day match-quality score improved from 0.675 to 0.739 - a genuine gain. But the average price gap to Brainpool got worse (-11.86 to -30.39 EUR/MWh), because the other side of the border had no storage or flexibility of its own yet, and the border limit itself was still a placeholder guess rather than a real number.

4. Real Border Data Replaces the Guess

Swapped the placeholder for a real, sourced border-capacity number.

Used the real cross-border flow data (once fully recovered) to work out a genuine, grounded limit on how much power can flow between Germany and its neighbours - replacing the placeholder guess with real numbers (roughly 21,600-22,000 MW each direction). This confirmed the earlier match-quality improvement was real and not just an artefact of an oversized placeholder.

5. Adding Real Storage and Subsidies - Our Best Result Yet

Gave the other side of the border its own real storage and subsidy treatment, the same way Germany already has.

Added genuine battery-style and reservoir hydro storage to the combined European zone, sized using real published hydro-industry figures, plus real government-subsidy treatment instead of a placeholder zero. Along the way, found and fixed three genuine bugs in how the two zones were wired together - including one where two zones sharing a single forecasting agent caused the model to crash outright, and another where the wrong kind of market agent was assigned a subsidy type it couldn't actually handle. Checked directly that the new storage was doing real work, not just sitting idle.

Result: average price within 60 cents of the real figure - the best match this whole project has produced.

Average price gap improved to just -0.59 EUR/MWh (was -30.39 two versions ago), with match quality holding strong and the number of 'ran out of power' hours dropping from 26 back down to a much more reasonable 7.

In plain terms: The first version was like two countries linked by a single wire with no batteries on either end - useful, but jumpy. Adding real storage on both sides smoothed that out, the same way home batteries smooth out a household's own solar swings.

6. Testing 2028 and 2029 - and Catching a Misleading Number

Extended the same build to the two other years - match quality held up strongly in both.

Applied the exact same build to 2028 and 2029, keeping the European side's data frozen at its real 2023 shape (the same principle already proven for Germany's own out-of-sample testing) so nothing was re-tuned per year. Match quality jumped strongly in both years (2028: 0.446 to 0.718, 2029: 0.446 to 0.752), confirming the earlier win wasn't a one-year fluke.

You asked: "What is next? Also, the bias for 2029 seems worse in the coupled result."

You were right to flag it - checked properly, and it was a misleading headline number, not a real regression.

The plain average price gap for 2029 did look concerning at first glance. Broke it down properly rather than accepting the headline number: found that 88% of that gap came from just a handful of extreme 'ran out of power' hours, not from ordinary days. On ordinary days - which is over 99% of the year - all three years are excellent and remarkably consistent (2027: -0.59, 2028: +1.08, 2029: +3.21 EUR/MWh). Corrected the Progress Report in place rather than leaving the misleading version standing, since letting an all-hours-only number stand would have understated a genuinely strong result.

7. Chasing the Last Gap: Two Honest Dead Ends

The one real, remaining pattern: those rare extreme hours get more frequent the further a year is from 2027 (7 in 2027, 17 in 2028, 76 in 2029). Two reasonable explanations were tested directly, one after the other.

You asked: "Test scaling ROE/transmission to keep pace with DE's growth."

Did not work - tested directly, and it made one year worse, not better.

The idea: Germany's own demand keeps growing while the border-capacity number stays flat, so scale the border capacity up to match. Built and ran it. Result: 2028 actually got worse (17 extreme hours became 25), and 2029 barely improved (76 became 68). Dug into why: checked hour-by-hour whether Germany was actually hitting its border limit during these extreme hours, and found it wasn't - the ceiling was only actually being hit in a small minority of cases, meaning capacity was never really the true bottleneck.

You asked: "Test the MaximumShiftedEnergyPerIterationInMWH fix."

Also wrong - opened up AMIRIS's own real program code to check, and said so plainly.

The next suspect: a specific internal setting that limits how much power AMIRIS's own matching engine can shift per round. Opened the actual compiled program (not just our settings file) to check its real default value directly - and found it's already set to effectively unlimited, not a hidden cap at all. This disproved the very hypothesis raised just beforehand, and that was stated directly rather than glossed over. Found one real, unexplained clue instead while investigating: Germany's import amount during these rare hours stays suspiciously flat no matter how much spare capacity is available - pointing to the matching engine's own internal stopping logic as the real limiter. Confirming that fully would mean editing and rebuilding AMIRIS's own program, a bigger step not yet taken.

In plain terms: Two sensible theories, tested properly instead of assumed - both ruled out with real evidence, leaving one genuine clue worth chasing next, rather than an unexplained shrug.

8. Bringing It All Together

You asked: "Update the supervisor deck with everything since Phase 34. / Update the progress report PDF with Phases 34-40 too."

Both documents fully updated with this session's work.

Added nine new slides to the supervisor deck, placed right after the earlier small-idea tests and before the closing summary slides - covering the build decision, the real data sourcing story, each of the four model versions, the 2028/2029 results (the corrected, accurate version), and the two honestly-ruled-out dead ends. Regenerated the technical Progress Report the same way, which already contained the full written record of every phase from this session.

	2027	2028	2029
Match quality (with two-way trading)	0.717	0.718	0.752
Average price gap, ordinary days	-0.59	+1.08	+3.21
Rare extreme hours (goal: near zero)	7	17	76

That last row is the one honest, open thread left for the next session: a real, narrow, well-diagnosed gap, not a mystery - with a solid next lead (the matching engine's own internal stopping logic) and a clear next step (patching and rebuilding AMIRIS itself) already identified.

What We Did This Session

A simple, no-jargon walkthrough of this session's conversation - what was asked, and what happened

6 September 2026

Two parts today. First, properly saving all our work under real version control for the first time - and catching a real security slip along the way. Second, three more real experiments chasing the very last open question on the cross-border trading work: a storage test, a single-country test, and a full 10-country build that hit and fixed a genuine software bug in AMIRIS itself.

1. Setting Up Real Version Control - and Catching a Real Slip

You asked: "Commit these changes."

Found the project had never actually been under its own version control - and stopped before committing a real credential by mistake.

Checking first (rather than just running the commit) turned up two real problems: the version-control history on this computer covered the ENTIRE home folder, not just this project, mixing in unrelated personal files - and partway through preparing the commit, spotted that the file holding our real API access tokens (ENTSO-E, renewables.ninja) was about to be saved into that history by mistake. Stopped, excluded it properly, and set up a clean, separate, dedicated history just for this project instead - the safer choice given what the first option risked.

Fixed the author details too, once flagged.

The save was initially credited to the wrong name/email (a leftover default on this shared computer). Corrected it to the right one and confirmed the save now shows the right authorship - since this was the very first save in a brand new history, fixing it in place was clean and safe, with nothing else to disturb.

Clarified after a follow-up question: this save exists only on this computer for now, not on GitHub - nothing has been uploaded anywhere yet. That's a separate, deliberate next step whenever you're ready for it.

2. One More Lead on the Last Open Question

You asked: "Continue building out the export module improvements."

Tested directly whether the other side's own storage was the real limiter - and ruled it out.

The one open thread left: a small number of hours where Germany imports less than it could, even with room to spare. Checked directly whether the rest of Europe's own batteries and reservoirs were simply empty during those hours. They weren't - neither ever came close to its own limit, in either year checked. Rules out 'nothing left to give' as the explanation, and points more firmly at the trading engine's own internal logic as the real cause - not yet fully provable without a deeper software change.

3. Pushing Further: One Real Country Instead of the Combined Zone

You asked: "What of building for each of the countries, maybe we should try that?"

Flagged the real risk first, then built a single-country test as a fair first check.

Warned directly that splitting every country out separately, the same simple way, could make things WORSE rather than better, based on what the earlier work had already shown. You agreed a single-country test first was the sensible way to check that before committing to the bigger build. Built France properly as its own real zone - the largest economy involved, with enough real data already on hand to do it right.

Result: a real step backward, with a clear, understood reason.

The 'ran out of power' hours jumped sharply (7 to 167) once France was split out. Checked why rather than leaving it a mystery: France's own huge nuclear fleet had been quietly propping up the WHOLE combined zone's supply. Pulled out on its own, the remaining combined zone lost that cushion, while France's own leftover cheap power could no longer reach it either, stuck behind a single narrow border link to Germany.

4. Going All the Way: All 10 Real Countries at Once

You asked: "Keep digging into the Denmark bug first."

Built all 10 real neighbouring countries as their own zones - and hit a genuine software bug.

Since one country made things worse for a clear reason, the natural next test was whether the SAME problem compounds across all 10, or whether something different happens at full scale. The build itself crashed - and this time, unlike every earlier problem in this project, the cause wasn't obvious from checking the settings.

Tracked it down properly rather than guessing: isolated it to Denmark specifically.

Tested one country at a time. Austria worked perfectly. Denmark, tested the same way, crashed every time. Checked Denmark's real numbers: its actual hydropower is only 7 megawatts - thousands of times smaller than every other country involved. Proved this was the real cause, not a coincidence, by testing it twice in each direction: the exact same setup with just the numbers made bigger ran perfectly, every time; the exact same setup with the real tiny numbers crashed, every time. Also tried simply leaving Denmark's hydropower out entirely - that did NOT fix it either, so it had to be properly patched, not just removed.

Fixed it honestly: a clearly labelled technical workaround, not invented data.

Applied a documented, openly-artificial minimum size to Denmark's (and the similarly-tiny Netherlands') hydropower, purely so the software could run - clearly marked in the file itself as NOT representing either country's real hydropower scale. With that fix in place, the full 10-country build ran cleanly all the way through.

	Combined zone (best)	France alone	All 10 countries
'Ran out of power' hours	7	167	120
Average price gap, ordinary days	-2.90	+9.65	+4.87
Match quality (trustworthiness)	0.717	0.716	0.731 (best yet)

Result: better than expected, but still not the winning approach.

Genuinely surprising: going all the way to 10 real countries did BETTER than splitting out just one - likely because Germany can now draw on many countries' cheap power at once, instead of being stuck behind a single narrow link. But it still falls short of the original combined-zone approach on the two measures that matter most. Decision: keep the combined zone as our best, standing result - both new builds are kept as real, documented evidence, not deleted.

5. Bringing Both Documents Up to Date

You asked: "Update the supervisor deck with Phase 43. / Update the daily session-recap PDF too."

Both the supervisor deck and the technical progress report now reflect all of today's real work.

Added five new slides to the supervisor deck covering the storage test, the France result, the Denmark bug story, and the final 10-country comparison table - placed right after the earlier market-coupling section and before the closing summary slides. The technical progress report already had the full written record of all three experiments from earlier in the session.

What We Did This Session

A simple, no-jargon walkthrough of this session's conversation - what was asked, and what happened
7 September 2026

A short, focused session: getting yesterday's local save-history actually onto GitHub, where it can be shared or accessed from anywhere. Three separate real technical snags came up along the way - each one found, explained, and fixed rather than worked around.

1. Getting Ready to Push

You asked: "Push it to GitHub."

Checked what was needed first, rather than just attempting it blindly.

This computer had no direct way to create a new project on GitHub by itself, so a real empty destination was needed first. Checked the total size of everything to be uploaded (85 megabytes - comfortably fine) before asking you to create a blank project space on your GitHub account and share its address.

You provided the address: `github.com/m-ajiboy/Masters-RES-Project`.

2. Snag One: A Broken Security Certificate Setting

The very first upload attempt failed immediately with a security-certificate error.

The version of the upload tool installed on this computer was pointing at a security certificate file in a location that didn't actually exist here - a leftover setting from a different, unrelated program. Found the REAL certificate file's actual location on this computer and pointed the setting there instead, just for this one project (not changed machine-wide, to avoid affecting anything else on this shared computer).

3. Snag Two: GitHub No Longer Accepts a Plain Password

The second attempt reached GitHub successfully - but GitHub itself rejected it.

GitHub stopped accepting an ordinary username-and-password login for this kind of upload some time ago, for security reasons - it now requires either a special access code or a different, more secure connection method entirely. Explained the two real options and asked which you'd prefer.

You asked: "git@github.com:m-ajiboy/Masters-RES-Project.git"

You chose the more secure connection method (known as SSH) and confirmed the exact address to use.

4. Snag Three: No Secure Key Existed on This Computer Yet

Switching connection methods surfaced one more real, expected step: this computer had never been set up for it before.

The secure method works using a matching pair of digital keys - one kept privately on this computer, one registered with your GitHub account. Neither existed here yet. First had to safely confirm GitHub's own identity (a standard, one-time check, similar to confirming you've reached the right website), then generated a brand new key pair and asked you to register the public half with your GitHub account.

You confirmed once the key had been added to your GitHub account.

5. Success: Everything Now Genuinely Saved on GitHub

Verified the new key worked, then completed the upload.

Checked directly that GitHub now recognised this computer before attempting the upload again, rather than assuming it would work. Both real saved versions of the project (the original full save, and yesterday's Phase 41-43 work) are now genuinely on GitHub at github.com/m-ajiboy/Masters-RES-Project - not just on this one computer any more.

In plain terms: Like moving from a locked filing cabinet in one office to a proper shared archive - the first attempt to move the files failed because of a broken address label, the second because the old-style door key GitHub used to accept isn't allowed any more, and the third because this particular office had never been issued the new-style key at all. Each problem was real, found, and fixed in turn - not skipped past.

One practical benefit going forward: the secure key generated today should make future uploads from this computer work smoothly, without needing to repeat any of today's setup.